# Royal Trees

Given $N$, $A$, $B$ and $D$, find number of trees of size $N$ in which $\text{dist}(A, B) = D$.

## Subtask 2

$D = 1$.

$$\#\text{trees} = (N-1)!$$

Since $B$ only has one choice

$$\#\text{trees} = \frac{(N-1)!}{B-1}$$

## Subtask 3

$A = 1, D = 2$.

Let $Q$ denote the node between $A$ and $B$.

$$\begin{aligned}\#\text{trees} &= \sum_{Q=2}^{B-1} \frac{(N-1)!}{(Q-1)(B-1)} \\ &= \frac{(N-1)!}{B-1} \sum_{Q=2}^{B-1} \frac{1}{Q-1}\end{aligned}$$

## Subtask 4

$D = 2$. In all subtasks, $\text{dist}(A, r) + \text{dist}(B, r) = D$ for a given node $r$. Cases:

- $\text{dist}(A, r) = \text{dist}(B, r) = 1$:

$$\begin{aligned}\#\text{trees}_1 &= \sum_{r=1}^{A-1} \frac{(N-1)!}{(A-1)(B-1)} \\ &= \frac{(N-1)!}{(A-1)(B-1)}(A-1) \\ &= \frac{(N-1)!}{B-1} \text{ unless } A = 1\end{aligned}$$

- $r = A$:

Let $Q$ denote the node between $A$ and $B$.

$$\begin{aligned}\#\text{trees}_2 &= \sum_{Q=A+1}^{B-1} \frac{(N-1)!}{(Q-1)(B-1)} \\ &= \frac{(N-1)!}{B-1} \sum_{Q=A+1}^{B-1} \frac{1}{Q-1}\end{aligned}$$

So,

$$\begin{aligned}\#\text{trees} &= \frac{(N-1)!}{B-1} + \left( \frac{(N-1)!}{B-1} \sum_{Q=A-1}^{B-1} \frac{1}{Q-1} \right) \\ &= \frac{(N-1)!}{B-1} \left[ \{1 \text{ if } A \neq 1\} + \sum_{Q=A+1}^{B-1} \frac{1}{Q-1} \right]\end{aligned}$$

## Subtask 5

$A = 1$.

Let the nodes $Q_1, Q_2, ..., Q_{k-1}, Q_k$ denote the $k = D - 1$ nodes between $A$ and $B$.

$$\#\text{trees} = \frac{(N-1)!}{B-1} \sum_{Q_1=2}^{B-k} \left( \frac{1}{Q_1 - 1} \sum_{Q_2=Q_1+1}^{B-k+1} \left( \frac{1}{Q_2 - 1} \sum_{Q_3=Q_2+1}^{B-k+2} \left( \frac{1}{Q_3 - 1} \cdots \right) \right) \right)$$

Let

$$f(i, j) = \sum_{Q=j+1}^{B-k+i} \frac{1}{Q-1} f(i+1, Q)$$

$$f(k-1, j) = \sum_{Q=j+1}^{B-1} \frac{1}{Q-1}$$

$$f(i, j) = f(i, j+1) + \frac{1}{j} f(i+1, j+1)$$

$$\text{if } j > B - D + i \text{ then } f(i, j) = 0$$

Therefore $\#\text{trees} = \frac{(N-1)!}{B-1} f(0, 1)$.

## Full

Let $r$ be a node such that $A, B \in \text{Desc}(r)$ and $\text{dist}(A, r) + \text{dist}(B, r) = D$.

Define $c(i)$ as:

$$c(i) = \begin{cases} 2 \text{ if } i < A \\ 1 \text{ if } A < i < B \\ 0 \text{ if } i = A \text{ or } i \geq B \end{cases}$$

Define $g(i, j)$ as where $k = D - 2$:

$$g(i, j) = \sum_{Q=j+1}^{B-k+i} \frac{c(Q)}{Q-1} g(i+1, Q)$$

$$g(k-1, j) = \sum_{Q=j+1}^{B-1} \frac{c(Q)}{Q-1}$$

$$g(k, j) = 1$$

$$\text{if } i < k \text{ then } g(i, B-1) = 0$$

$$g(i, j) = g(i, j+1) + \frac{c(j+1)}{j} g(i+1, j+1)$$

$$\#\text{trees} = \frac{(N-1)!}{B-1} f(0, 1) \qquad [A = 1]$$

$$\#\text{trees} = \frac{(N-1)!}{(A-1)(B-1)} \sum_{r=1}^{A-1} g(0, r) + \frac{(N-1)!}{B-1} f_{D-1}(0, A) \qquad [A > 1]$$